

Deon Levon Dmello

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EXPERIENCE

AI Engineer I

April 2026 – Present

Bayer

Bengaluru, KA

- Developed AI agents and agentic workflows using LLMs and RAG pipelines that reduced approval processes from days to minutes, driving significant operational efficiency across business units.
- Designed and implemented backend services and REST APIs to support agent orchestration and enterprise data integration.
- Led API testing and quality assurance efforts for AI-driven systems, ensuring reliability and consistency of agentic workflows in production environments.

AI Engineer

June 2024 – March 2026

Tardid Technologies

Bengaluru, KA

- **LiDAR–Camera Calibration:** Developed an automated tool for extrinsic calibration between LiDAR and camera sensors, improving spatial alignment accuracy and reducing manual calibration effort.
- **Real-time & Zero-Shot Action Recognition:** Developed and deployed action recognition systems using X3D, Swin3D, and X-CLIP for zero-shot inference, achieving 30 FPS on live RTSP streams.
- **Vision-Language Models (VLMs):** Integrated CLIP and X-CLIP based pipelines for cross-modal image–text retrieval, zero-shot classification, and scene understanding.
- **Large Language Models (LLMs):** Deployed and optimized instruction-tuned LLMs for vision-assisted analytics and system control using llama.cpp and SGL, enabling low-latency inference on edge GPUs.
- **Face Recognition & Tracking:** Implemented a perimeter surveillance system using face embeddings stored in a vector database with cosine similarity for identity matching, integrated with OC-SORT for multi-person tracking.
- **Machine Learning & Deep Learning:** Fine-tuned CNN architectures for object classification, achieving a 10% improvement in F1-score on real-world datasets.
- **Backend Development:** Designed and maintained FastAPI-based backend APIs for ML inference, model orchestration, and data pipelines.
- **Infrastructure & Version Control:** Administered a private Git (Gitea) instance, managing CI workflows and onboarding 10+ developers.

SRE / Platform Engineer Intern

March 2024 – June 2024

Practo Technologies

Bengaluru, KA

- Managed and resolved support tickets using JIRA, connecting to servers over SSH to troubleshoot and fix issues, ensuring system reliability and uptime.
- Implemented bug patches, enhanced platform stability, and developed a thorough understanding of in-house tools for sending SMS and email notifications.
- Managed backend microservices using Docker, ensuring efficient deployment, scalability, and reliability of platform services.
- Performed scheduled database backups and managed production Postgres dumps to support data recovery and environment migrations.
- Developed a data pipeline to collect and analyze logs from AWS CloudWatch, enabling proactive identification of errors and deployment issues.

EDUCATION

Indian Institute of Technology Madras

Chennai, TN

BS in Programming and Data Science

September 2021 – Present

- Completed Diploma in Data Science & Diploma in Programming

Reva University

Bengaluru, KA

B.Tech Computer Engineering (AI & ML)

December 2020 – July 2024

PROJECTS

Sync'em | *FastAPI, Vue.js, TailwindCSS, PostgreSQL, Celery, Redis, GenAI Agents, Jira* | GitHub September 2025

- Built a full-stack workforce and project management platform in an Agile Scrum environment with a cross-functional 8-member team, delivering role-based portals for Admins, HR, Project Managers, and Employees.
- Implemented Generative AI assistants (LLM + RAG) to automate HR queries, onboarding workflows, and requirement clarification, significantly reducing manual support effort.
- Integrated asynchronous job pipelines with Celery and Redis for automated email notifications, scheduled reports, and background processing.
- Documented APIs using OpenAPI (Swagger) and wrote Pytest test suites for critical services, improving system reliability.

QuizNexus | *Vue.js, Flask, SQLite, Bootstrap, JWT, Redis, LangChain, LangGraph* | GitHub July 2025

- Developed a full-stack quiz platform with a Flask REST API and modular Vue.js frontend for dynamic quiz creation, administration, and responsive UI design.
- Built an AI-powered student advisor using LangChain and LangGraph to analyze quiz history, detect weak areas, and generate personalized study recommendations.
- Implemented secure role-based authentication using JWT, Redis caching for session optimization, and MailHog/SMTP for automated email verification.

ImageColorization | *Python, PyTorch, CNN, UNet, GANs* | GitHub April 2024

- Developed a CNN-based model using PyTorch to automatically colorize grayscale images, applying UNet, GANs, convolutional layers, and loss optimization techniques.
- Packaged the model into a Python module with clear documentation; extending into a CLI tool for enhanced usability and portability.

TECHNICAL SKILLS

Languages: Python, C/C++, Bash, SQL, Go

Databases & Storage: PostgreSQL, Redis, Object Storage

Machine Learning & Deep Learning: scikit-learn, PyTorch, OpenCV, CLIP

GenAI & RAG Systems: LangChain, LangGraph, CrewAI, HuggingFace

LLM Inference & Optimization: llama.cpp, SGL, Quantized LLMs, RAG Pipelines

Vector Stores & Embeddings: FAISS, Qdrant

MLOps & Model Serving: FastAPI, REST APIs, Model Orchestration, CI/CD, Edge Deployment, GPU Inference

Cloud & Reliability: AWS (EC2, CloudWatch), PagerDuty, Jira, SSH, Linux Server Administration

Developer Tools: Docker, Git/Gitea, VS Code, Jupyter, Colab, Neovim

Operating Systems: Linux (Ubuntu), Windows

CERTIFICATIONS

- Advance Certificate in Programming and Application Development — IIT Madras
- Foundational Level Certificate in Programming and Data Science — IIT Madras
- TensorFlow Developer Certificate
- The Web Developer Bootcamp (MERN)
- Python for Data Science, AI & Development
- Programming for Everybody (Python)
- Design Thinking for Innovation